6. Adversary Emulation Practice

1. Objective:

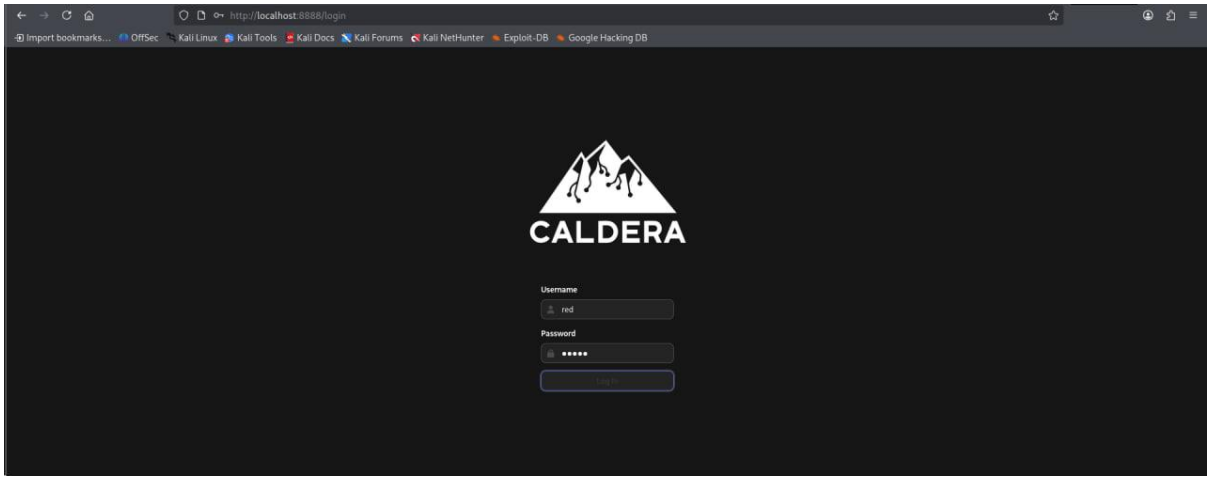
To simulate adversary techniques using MITRE Caldera and evaluate SOC detection capabilities through Wazuh monitoring.

2. Tools Used:

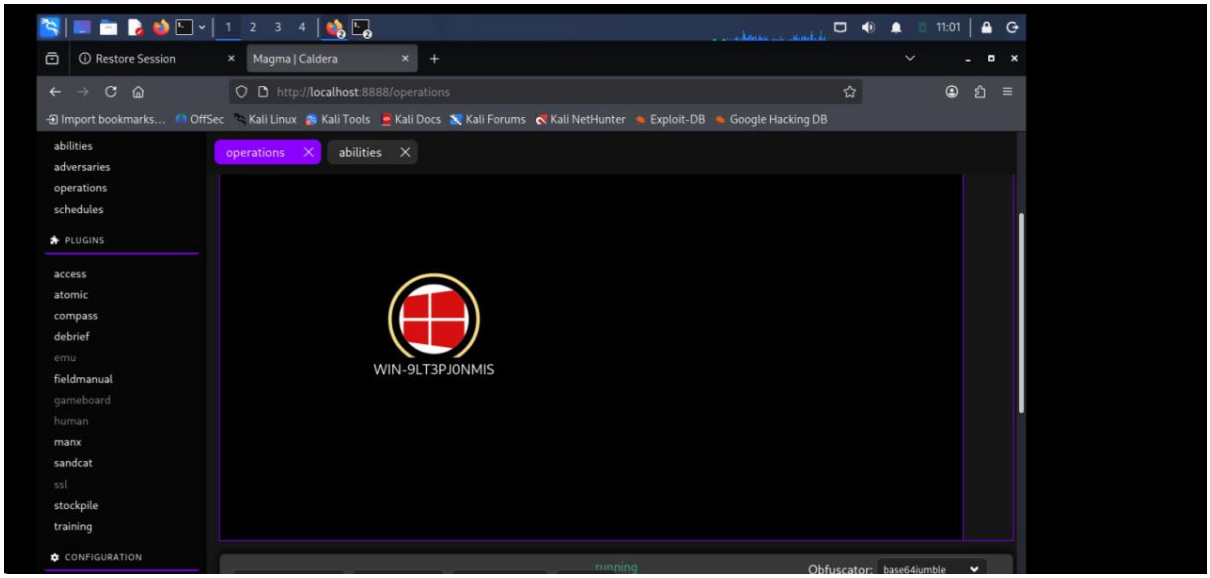
MITRE Caldera
Wazuh SIEM
Windows 8.1 (Victim Machine)
Kali Linux (Attacker Machine)

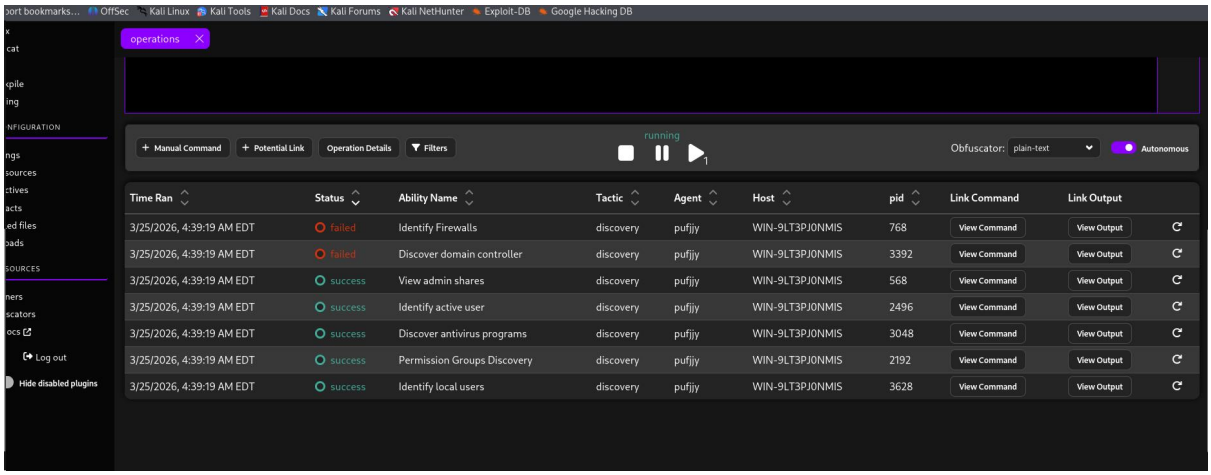
3. Attack Simulation (T1566 – Spearphishing):

An adversary emulation operation was executed using MITRE Caldera to simulate a spearphishing attack (MITRE ATT&CK ID: T1566).



The attack involved executing command-based payloads on the Windows victim machine to mimic user interaction with a malicious link or file.





The Caldera agent executed the following activities:

User identification (whoami)

System information gathering

Network enumeration

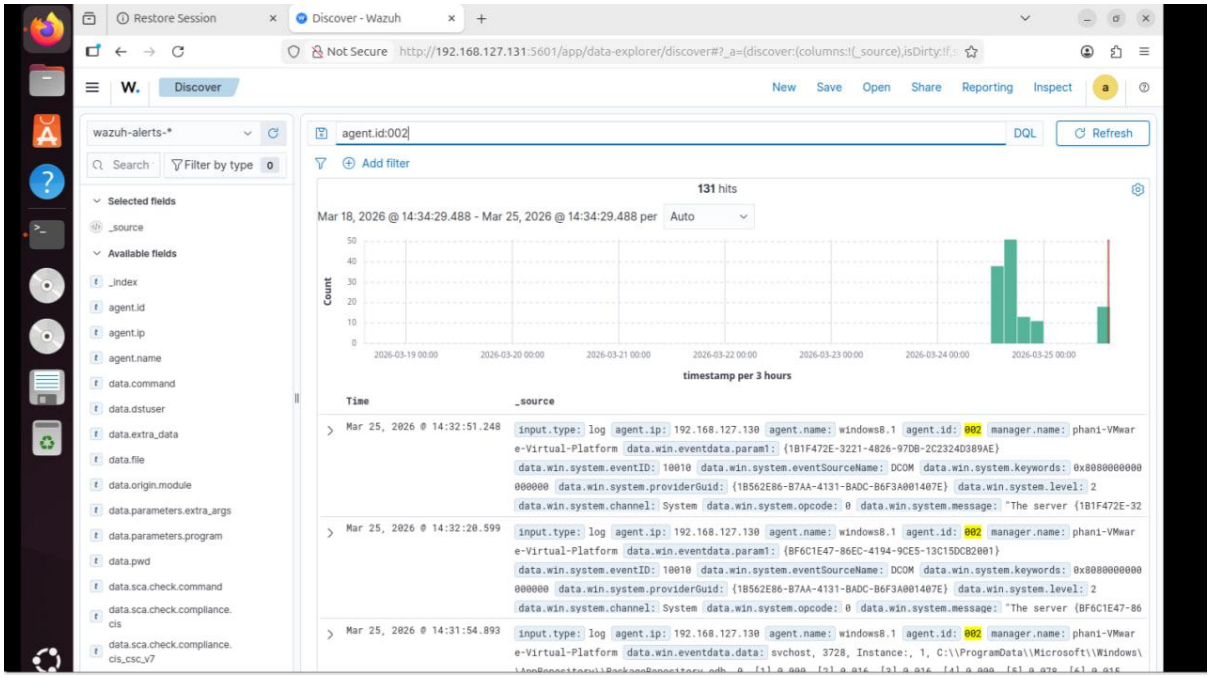
4. **Detection Configuration (Wazuh)**:

Wazuh was configured to monitor endpoint activity through the installed agent on the Windows machine. Logs related to process creation, command execution, and user activity were collected and analyzed.

Detection was verified by:

Monitoring Security Events → Discover

Filtering logs using agent.name and command-based queries
Checking alerts in /var/ossec/logs/alerts/alerts.log



5. Emulation Results:

Timesstamp	TTP	Detection Status	Notes
3/25/2026 14:32	T1566	Detected	Phishing activity simulated; command execution observed |

6. Analysis:

1. The Caldera operation executed successfully on the victim machine.
2. Wazuh collected logs corresponding to command execution activities.
3. Detection was achieved through process monitoring rather than direct phishing detection.

7. Detection Gaps Identified:

- - No direct email-based phishing detection.
- - Lack of behavioral correlation rules.
- - Absence of Sysmon reduces detailed visibility.

8. Emulation Report (100 Words):

The adversary emulation using MITRE Caldera successfully simulated a spearphishing attack (T1566) by executing commands on the target Windows system. Wazuh detected the activity through process monitoring and log analysis, confirming that endpoint-level visibility is functioning correctly. However, the detection was limited to command execution and did not directly identify phishing behavior. This highlights a gap in email-based threat detection and behavioral correlation. Additionally, the absence of Sysmon integration reduced detailed telemetry visibility. To improve detection capability, advanced rule tuning, Sysmon deployment, and integration